

Fig. 2: Power supply circuit for the wave generator

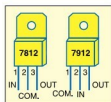


Fig. 3: Pin configurations of ICs 7812 and 7912

and A2 (IC1). Op-amp A1 acts as a standard squarewave generator and A2 acts as an integrator or ramp generator with the slope determined by the current flowing into pin 1 of A2, thus allowing voltage control.

Op-amp A1 operates as a Schmitt trigger with its trigger levels centered about the ground line. The timing capacitor is selectable for different frequency ranges by rotary switch S2. The triangular and square waves with 1V peak-to-peak level are available from pins 8 and 9 of IC1, respectively. These are fed to waveform-selector switch S3 through level-matching resistors R5 and R7. Due to the buffer stage in op-amp A1, the squarewave output has an offset voltage. Resistor R6 adds a small current to correct this.

The triangular output is also fed to op-amp A3 via resistor R8. Op-amp A3 is configured as a voltage-controlled filter to track the oscillation frequency. It filters the triangular wave to give a fairly good sine wave.

The filter is voltage-controlled in parallel with the oscillator, so the ratio of input frequency to the filter's cut-off frequency is constant. This ensures that the sine wave amplitude is the same at all frequencies. The resulting sine wave is fed to wave-selector switch S3 via resistor R11. Preset VR3 is an offset-null controller.

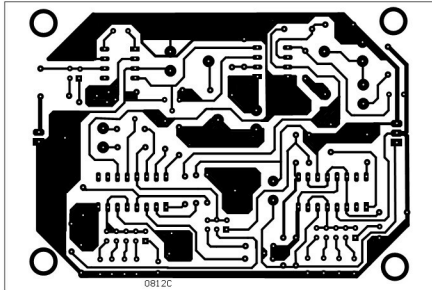


Fig. 4: An actual-size, single-side PCB for the wave generator circuit

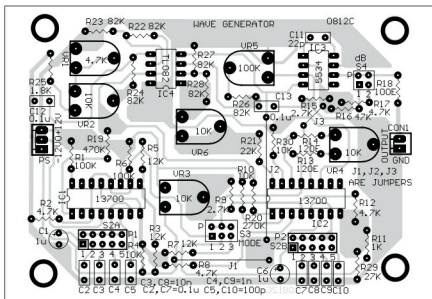


Fig. 5: Component layout for the PCB in Fig. 4

From switch S3, the chosen output is fed to op-amp A4, which is a fairly conventional voltage-controlled amplifier. It gives an output in the form of a current, which has switchable gain via switch S4 to allow fine control at low output levels.

The control voltages for the voltage-controlled oscillator/filter and the voltage-controlled amplifier swing both positive and negative with respect to ground. Since it is convenient that any external control signal be purely positive going, IC4 (A5 and A6) pro-